

# Functional Analysis

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The foundational knowledge relating to Measure & Ergodic Theory, that has not been covered elsewhere, was provided by Jamneshan and Kreidler (2025).

**Definition 0.1.** For a Hilbert space  $H$ ,

1. A *linear isometry* is a linear map  $U : H \rightarrow H$  such that  $\|Uf\| = \|f\|$  for all  $f \in H$  (and thus injective).
2. A linear isometry,  $U$ , is a *unitary operator* if it is surjective (and thus bijective).
3. We write  $\mathcal{U}(H)$  for the set of all unitary operators  $U : H \rightarrow H$ .
4. For a group  $\Gamma$ , we call a group homomorphism  $U : \Gamma \rightarrow \mathcal{U}(H)$ ,  $\gamma \mapsto U_\gamma$  a *unitary representation* of  $\Gamma$  on  $H$ .

**Definition 0.2.** Let  $J : (Y, S) \rightarrow (X, T)$  be an extension of measure-preserving systems. Then  $E = U_J(L^2(Y))$  is an *invariant Markov sub-lattice* of  $L^2(X)$ , i.e.,

1.  $E$  is a closed linear subspace of  $L^2(X)$ ,
2.  $1 \in E$ ,
3.  $|f|, \operatorname{Re}(f), \operatorname{Im}(f) \in E$  for every  $f \in E$ , and
4.  $U_{T_\gamma} f \in E$  for every  $f \in E$  and  $\gamma \in \Gamma$ .

**Definition 0.3.** Let  $H$  be a Hilbert space.  $\mathcal{L}(H)$  is the space of all bounded linear operators from  $H$  to  $H$ .<sup>1</sup>

A family  $\mathcal{S} \subseteq \mathcal{L}(H)$  is a *semigroup (of operators)* if  $UV \in \mathcal{S}$  for all  $U, V \in \mathcal{S}$ . It is a *contraction semigroup* if, in addition,  $\|U\| \leq 1$  for all  $U \in \mathcal{S}$ .

We call

$$\operatorname{fix}(\mathcal{S}) := \bigcap_{U \in \mathcal{S}} \operatorname{fix}(U) = \{f \in H \mid Uf = f \text{ for every } U \in \mathcal{S}\}$$

the *fixed space* of  $\mathcal{S}$ .<sup>23</sup>

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<sup>1</sup>Note,  $\mathcal{U}(H) \subseteq \mathcal{L}(H)$  and  $\|U\| = 1$  when  $U \in \mathcal{U}(H)$ , so  $\mathcal{U}(H)$  is a contraction semigroup.

<sup>2</sup> $\operatorname{fix}(\mathcal{S}) \subseteq H_{\text{ds}}$ .

<sup>3</sup>Meeting Notes: 0-dimensional. Next most simple case would be scaling (1-dimensional), and then rotations on a plane (2-dimensional).

The *closed convex hull*  $\overline{\text{co}} A$  is the closure of the set of all convex combinations of elements of  $A$ .

The *closed linear hull*  $\overline{\text{lin}} A$  is the closure of the set of all linear combinations of elements of  $A$ .

**Definition 0.4.** We write  $\mathbb{T} := \{z \in \mathbb{C} \mid |z| = 1\}$  for the multiplicative group of complex numbers of modulus one, and call this the *torus*.

**Definition 0.5.** Let  $U : \Gamma \rightarrow \mathcal{U}(H)$  be a unitary representation of a discrete abelian amenable group  $\Gamma$ .

A group homomorphism  $\chi : \Gamma \rightarrow \mathbb{T}$ , where  $\mathbb{T} = \{z \in \mathbb{C} \mid |z| = 1\}$ , is called a *character*. The *dual group*  $\Gamma^*$  of  $\Gamma$  is the set of all such characters equipped with the multiplication given by  $(\chi_1 \chi_2)(\gamma) := \chi_1(\gamma) \chi_2(\gamma)$  for  $\gamma \in \Gamma$  and  $\chi_1, \chi_2 \in \Gamma^*$ .<sup>4</sup>

**Definition 0.6.** Let

$$M_f := \text{lin}\{U_\gamma f \mid \gamma \in \Gamma\}.$$

We call a subset  $M \subseteq H$  an *invariant finite-dimensional subspace* if  $M_f$  is finite-dimensional and  $U_\gamma f \in M$  for all  $f \in M$  and  $\gamma \in \Gamma$ .

The closure

$$H_{\text{ds}} = \overline{\bigcup \{M \subseteq H \mid M \text{ invariant finite-dimensional subspace}\}} \subseteq H$$

is called the *discrete spectrum part* of  $U$ .<sup>5</sup>

**Definition 0.7** (Jamneshan and Kreidler, 2025, Definition 5.1.8). Let  $U : \Gamma \rightarrow \mathcal{U}(H)$  be a unitary representation of an abelian group  $\Gamma$ . For a character  $\chi \in \Gamma^*$ , we call

$$\begin{aligned} \ker(\chi - U) &:= \{f \in H \mid U_\gamma f = \chi(\gamma)f \text{ for every } \gamma \in \Gamma\} \\ &\bigcap_{\gamma \in \Gamma} \ker(\chi(\gamma)\text{Id}_H - U_\gamma) \end{aligned}$$

the *eigenspace* of  $U$  with respect to  $\chi$ , and elements  $f \in \ker(\chi - U) \setminus \{0\}$  are called *eigenvectors* with respect to  $\chi$ . The set  $\sigma_p(U) := \{\chi \in \Gamma^* \mid \ker(\chi - U) \neq \{0\}\}$  is the *point spectrum* of  $U$  and its elements are called the *eigenvalues* of  $U$ .

**Proposition 0.1** (cf. Jamneshan and Kreidler, 2025, Proposition 5.1.10). *The eigenspaces  $\ker(\chi - U)$  for  $\chi \in \Gamma^*$  are pairwise orthogonal. For  $M \subseteq H$ , then the following are equivalent:*

1.  *$M$  is an irreducible invariant finite-dimensional subspace.*
2.  *$M$  is an invariant linear subspace which is at most one-dimensional.*
3. *There is  $\chi \in \Gamma^*$  and  $f \in \ker(\chi - U)$ , such that  $M = \mathbb{C} \cdot f$ .*

<sup>4</sup>Commutator subgroup and abelianisation?

<sup>5</sup>Meeting Notes: In the  $\Gamma = \mathbb{Z}$  case, then  $H_{\text{ds}}$  can be broken up into one-dimensional objects that corresponds to an eigenfunction.

**Definition 0.8.** A unitary representation  $U : \Gamma \rightarrow \mathcal{U}(H)$  of a topological group  $\Gamma$  is *strongly continuous* if  $\Gamma \rightarrow H, x \mapsto U_x f$  is continuous for every  $f \in H$ .

**Definition 0.9.** For every discrete group  $\Gamma$ , the maps

$$L : \Gamma \rightarrow \mathcal{U}(L^2(\Gamma)), \quad x \mapsto L_x \quad (1)$$

$$R : \Gamma \rightarrow \mathcal{U}(L^2(\Gamma)), \quad x \mapsto R_x \quad (2)$$

with  $L_x f := f \circ l_{x^{-1}}$  and  $R_x f := f \circ r_{x^{-1}}$  for  $f \in L^2(\Gamma)$  and  $x \in \Gamma$  are strongly continuous unitary representations. We call  $L$  and  $R$  the *left* and *right regular representation* of  $\Gamma$ , respectively.<sup>6</sup>

Assume that  $U : \Gamma \rightarrow \mathcal{U}(H)$  is a strongly continuous unitary representation of a discrete group  $\Gamma$ . Let further  $P$  be the orthogonal projection onto the fixed space  $\text{fix}(U(\Gamma))$ . Then

$$(Pf \mid g) = \lim_{N \rightarrow \infty} \frac{1}{|\Phi_N|} \sum_{\gamma \in \Phi_N} (U_\gamma f \mid g)$$

for all  $f, g \in H$ .

If  $U : \Gamma \rightarrow \mathcal{U}(H)$  is a strongly continuous unitary representation of a compact group  $\Gamma$ , then  $U$  has discrete spectrum.<sup>7</sup>

**Exercise 0.1.** Show that

$$H = H_{\text{ds}} \oplus H_{\text{wm}}$$

and what happens to  $H_{\text{ds}}$  and  $H_{\text{wm}}$  when it is averaged by the Abstract Mean Ergodic Theorem for amenable groups.<sup>8</sup>

**Proposition 0.2.** Let  $(X, T)$  be a measure-preserving system. Then  $L^2(X)_{\text{ds}}$  is an invariant Markov sublattice of  $L^2(X)$ .

**Exercise 0.2** (Jamneshan and Kreidler, 2025, Lemma 6.1.12). Assume that  $U : \Gamma \rightarrow \mathcal{U}(H)$  is a strongly continuous unitary representation of a discrete group  $\Gamma$ . Let further  $P$  be the orthogonal projection onto the fixed space  $\text{fix}(U(\Gamma))$ . Then  $(Pf \mid g) = \lim_{N \rightarrow \infty} \frac{1}{|\Phi_N|} \sum_{\gamma \in \Phi_N} (U_\gamma f \mid g)$  for all  $f, g \in H$ .

Jamneshan, A. and Kreidler, H. (2025). *ISem 28: Ergodic structure theory and applications*.

<sup>6</sup>As  $\Gamma$  is discrete, then any map  $\Gamma \rightarrow H$  is continuous for any topological space  $H$ . Thus,  $\Gamma \rightarrow L^2(\Gamma)$  defined by  $\gamma \mapsto L_\gamma f$  or  $\gamma \mapsto R_\gamma f$  is always continuous. Hence,  $L$  and  $R$  are strongly continuous unitary representations.

<sup>7</sup>This  $\Gamma$  would be the compact group behind the factor subsystem of the larger system. For example,  $\Gamma = \mathbb{Z}$  for the irrational  $\alpha$  group rotation would give us compact  $G = \{z \in \mathbb{C} \mid |z| = 1\}$ . The homomorphism from  $\Gamma \rightarrow G$  is  $n \mapsto e^{2\pi i \alpha n}$ .

<sup>8</sup>(Proposition 5.3.3)

$$H_{\text{wm}} = \left\{ f \in H \mid \lim_{N \rightarrow \infty} \frac{1}{|\Phi_N|} \sum_{\varphi \in \Phi_N} |(U_\varphi f \mid f)|^2 = 0 \right\}.$$